LECTURE NOTES

Lecture 13: Constrained Optimization

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Contents

1 Contrained Optimization

Minimizing an objective subject to design point restrictions called **constraints**

- New optimization problem statement

$$\min_x f(x)$$
$$\text{s.t. } x \in \mathcal{X}$$

Where $\mathcal{X}$ is the **feasible set**

- $\mathbb{R}^n$ or $\mathbb{Z}^m$ contrained to a subset

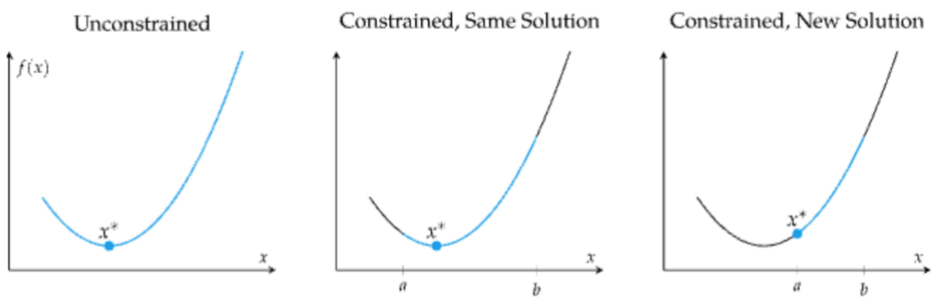

Figure 1: 1 dimention of constraints, where the solution changes

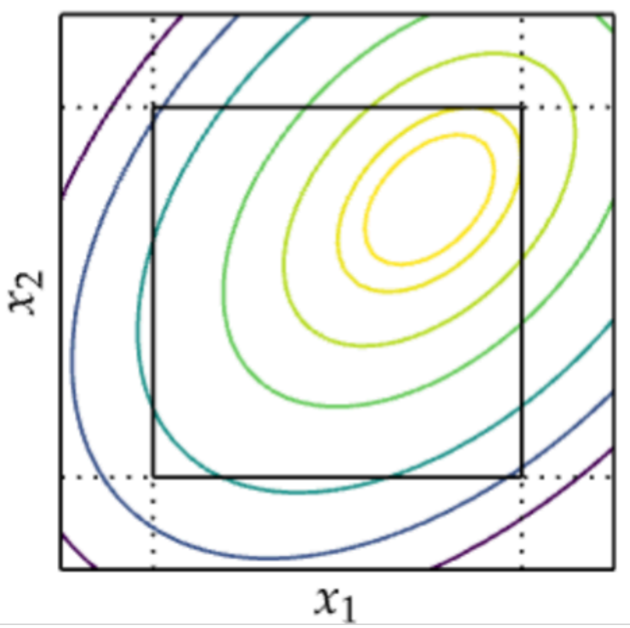

Figure 2: 2 dimentions

- Generally, constraints are formulated using two types:
  1) Equality constraints: $h(x) = 0$
  2) Inequality constraints $g(x) \leq 0$

2 Transformations to Remove Constraints

If necessary, some problems can be reformulated to use constraints in the objective function

Let x be constrained between a and b

$$x = t_{a,b}(\hat{x}) = \frac{b+a}{2} + \frac{b-a}{2}\left(\frac{2\hat{x}}{1+\hat{x}^2}\right)$$

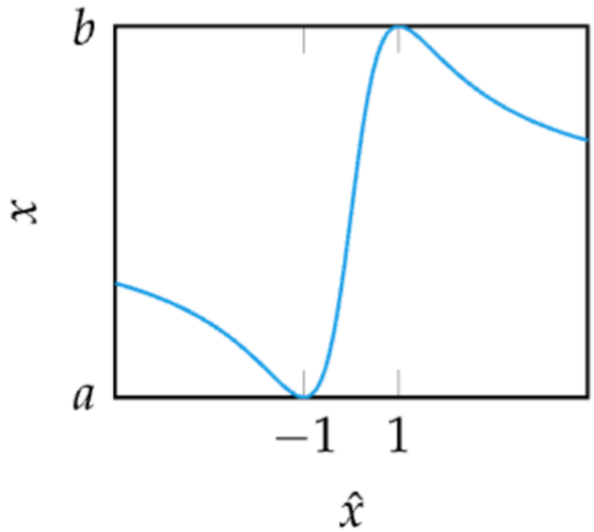

Figure 3: transformation

3 Lagrangian Multiplier Method

The method of Lagrangian Multipliers is used to optimize a function subject to (equality) constraints

(only equality constraints, critical points, because gradient of f and the gradient of h are aligned)

So instead of $\begin{array}{l} \min_x f(x) \\ \text{ s.t. } x \in \mathcal{X} \end{array}$

We can use the **Langrangian function**

- $\mathcal{L}(x, \lambda) = f(x) - \lambda h(x)$
- Then $\nabla_x \mathcal{L}(x, \lambda) = 0 \quad \text{and} \quad \nabla_\lambda \mathcal{L}(x, \lambda) = 0$
- This gets $\nabla f(x) = \lambda \nabla h(x) \qquad h(x) = 0$
- Then solve for x and λ

Intuitively, the method of Lagrange multipliers finds the point x^* where the constraint function is orthogonal to the gradient (since ∇h is normal to h)

Note: this only works wiht equallity constraints

3.1 Inequality Constraints

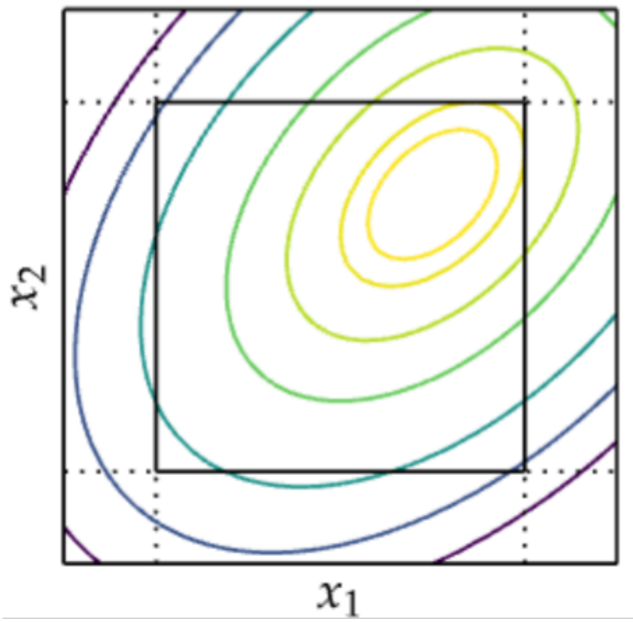

Figure 4: For inequality constraints, the local behavior fo the gradient in the exact meeting point of two constraint functions is a ned gradient that cannot be improved.

$$\nabla f(\vec{x}^*) = \lambda_1 \nabla g_1(\vec{x}^*) + \lambda_2 \nabla g_2(\vec{x}^*)$$
$$\lambda_1, \lambda_2 \geq 0$$

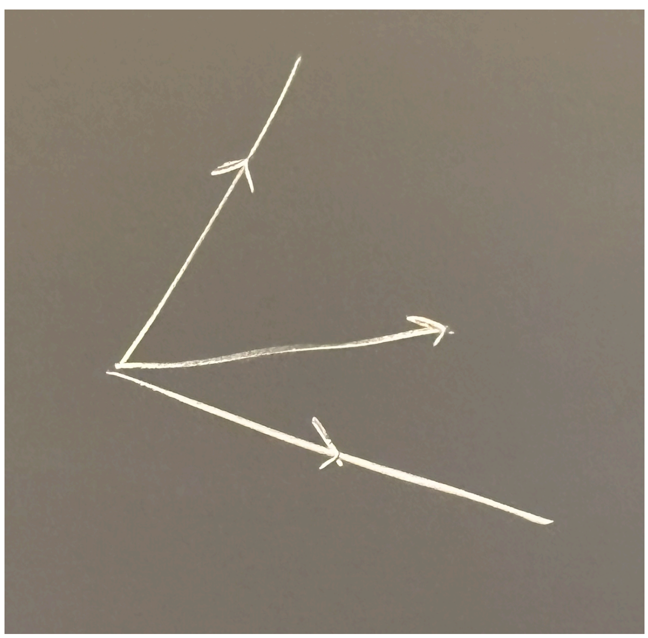

Figure 5: The new gradient vector composed of two gradients

Lets do the Langrangian Function Method

$$\mathcal{L}_\infty(x) = f(x) + \infty(g(x) > 0)$$

Though this is impractical because it is discontinuous and nondifferentiable.

- Instead, for $\mu \geq 0$:
  1) $\mathcal{L}(x, \mu) = f(x) + \mu g(x)$
  2) $\mathcal{L}_\infty(x) = \max_\mu \mathcal{L}(x, \mu)$
- For x infeasible $\mathcal{L}\infty(x) = \infty$; for x frasible $\mathcal{L}_\infty(x) = f(x)$
- The new optimization problem becomes

$$\min_x \max_{\mu \geq 0} \mathcal{L}(x, \mu)$$

This is called the **primal problem**

3.2 Necessary Conditions - KKT Conditions (Karush-Kuhn-Tucker)

For (FONC) x^* to be a critical point then we need

$$\begin{cases} \nabla f(\vec{x}^*) = \vec{\lambda} \nabla \vec{g}(\vec{x}^*) \\ \vec{g}(\vec{x}^*) \geq 0 \\ \vec{\lambda} \geq 0 \\ \vec{\lambda} \cdot \vec{g}(\vec{x}^*) = 0 \end{cases}$$

Particular cases:

- f is concave, g is convex, then KKT are also sufficient
- x^* interior point
- Patological cases where they do not hold (formal expressions of the conditions include assumptions to avoid these cases).
- Generalized Lagrangian Function:

$$\mathcal{L}(x, \mu, \lambda) = f(x) + \sum_i \mu_i g_i(x) + \sum_i \lambda_i h_i(x)$$

- The **primal form** of the optimization problem

$$\min_x \max_{\mu \geq 0, \lambda} \mathcal{L}(x, \mu, \lambda)$$

- Reversing the order of operations leads to the **dual form**

$$\max_{\mu \geq 0, \lambda} \min_x \mathcal{L}(x, \mu, \lambda)$$

Theorem (Max-min inequality)

For any function $f : Z \times W \to \mathbb{R}$,

$$\underbrace{\sup_{z \in Z} \inf_{w \in W} f(z, w)}_{\text{dual}} \leq \underbrace{\inf_{w \in W} \sup_{z \in Z} f(z, w)}_{\text{primal}}$$

For us:

$$\max_{\mu \geq 0, \lambda} \min_{x} \mathcal{L}(x, \mu, \lambda) \leq \min_{x} \max_{\mu \geq 0, \lambda} \mathcal{L}(x, \mu, \lambda)$$

Therefore, the solution to the dual problem d^* is a lower bound to the primal solution p^*

4 Penalty methods

Example

$$\begin{aligned} \underset{\mathbf{x}}{\text{minimize}} \quad & f(\mathbf{x}) \\ \text{subject to} \quad & \mathbf{g}(\mathbf{x}) \leq \mathbf{0} \\ & \mathbf{h}(\mathbf{x}) = \mathbf{0} \end{aligned}$$

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) + \rho \cdot p_{count}(\mathbf{x}) \\ \text{s.t.} \quad & p_{count}(\mathbf{x}) = \sum_i (g_i(\mathbf{x}) > 0) + \sum_j (h_j(\mathbf{x}) \neq 0) \end{aligned}$$

Figure 6: Penalty by counting violations

Procedure penalty_method;
Input: $f, p, \mathbf{x}, k_{max}; \rho = 1, \gamma = 2$
Output: $\mathbf{x}$ solution
for k in $1, \ldots, k_{max}$ **do**
  $\mathbf{x} \leftarrow \text{minimize}_{\mathbf{x}}\{f(\mathbf{x}) + \rho \cdot p(\mathbf{x})\}$;
  $\rho \leftarrow \rho \cdot \gamma$;
  if $p(\mathbf{x}) = 0$ **then**
    return $\mathbf{x}$;
return $\mathbf{x}$;

Figure 7: pentalty methods

- Count penalty:

$$p_{\text{count}}(x) = \left(\sum_i g_i(x) > 0\right) + \sum_j \left(h_j(x) \neq 0\right)$$

- Quadratic penalty:

$$p_{\text{quadratic}}(x) = \sum_i \max(g_i(x), o)^2 + \sum_j h_j(x)^2$$

- Mixed penalty:

$$p_{\text{mixed}}(x) = \rho_1 p_{\text{count}(x)} + \rho_2 p_{\text{quadratic}(x)}$$

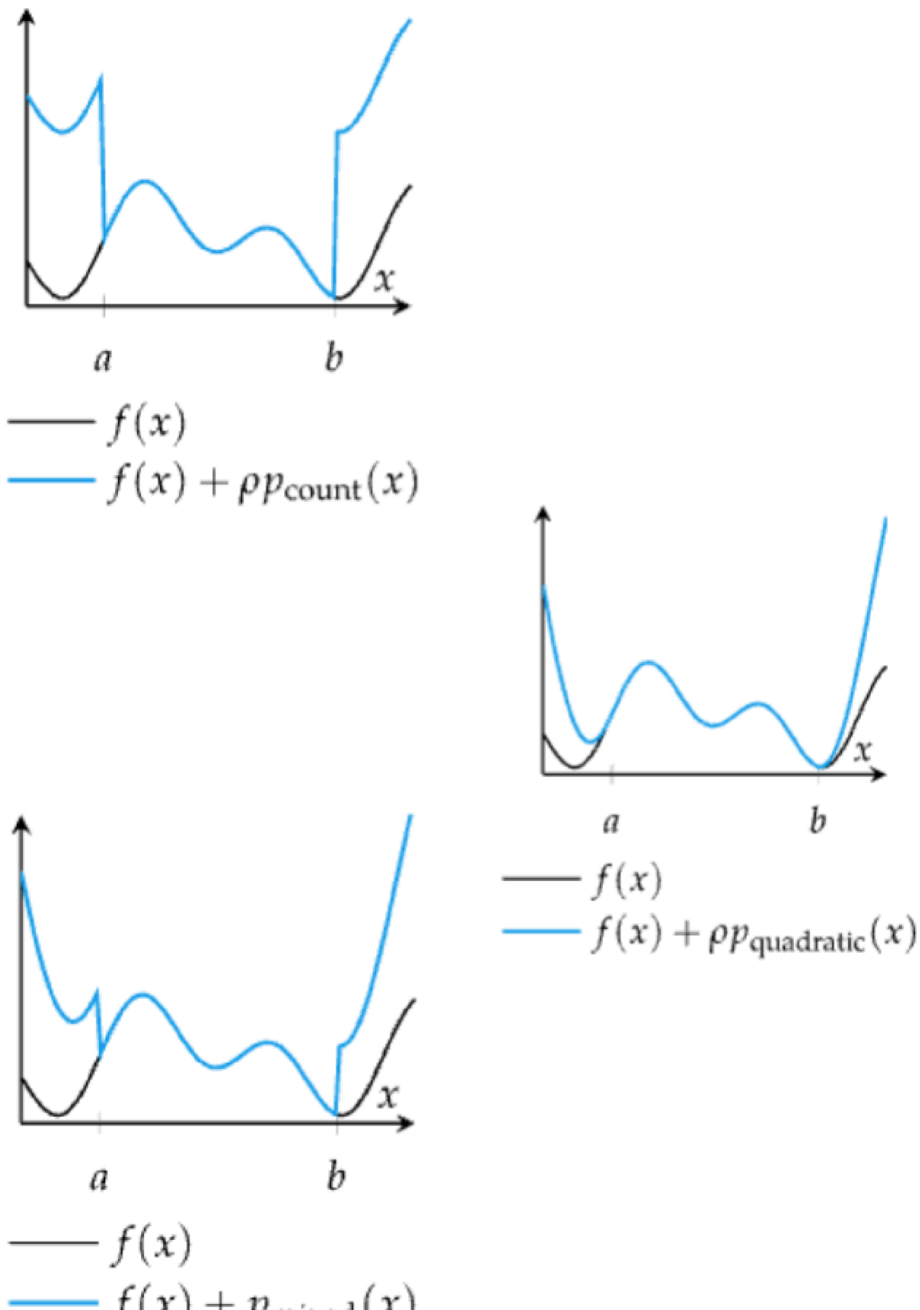

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Figure 8: multiple penalty methods

4.1 Augmented Lagrange Method

Procedure augmented_lagrange_method;
Input: $f, h, \mathbf{x}, k_{max}; \rho = 1, \gamma = 2)$
$\boldsymbol{\lambda} \leftarrow \mathbf{0}$;
for k in $1, \ldots, k_{max}$ **do**
  $p \leftarrow (x \mapsto \rho/2 \cdot \sum_i (h_i(\mathbf{x})^2) - \boldsymbol{\lambda} \cdot h(\mathbf{x}))$;
  $\mathbf{x} \leftarrow \text{minimize}_{\mathbf{x}}\{f(\mathbf{x}) + p(\mathbf{x})\}$;
  $\boldsymbol{\lambda} \leftarrow \boldsymbol{\lambda} - \rho \cdot \mathbf{h}(\mathbf{x})$;
  $\rho \leftarrow \rho \cdot \gamma$;
return $\mathbf{x}$;

Figure 9: Adaptation of penalty method for equality constraints

$$p_{\text{Lagrangian}}(x) \overset{\text{def}}{=} \frac{1}{2}\rho \sum_i (h_i(x))^2 - \sum_i \lambda h_i(x)$$

4.2 Interior Point Methods

Also called **barrier methods**, interior point methods ensure that each step is feasible

This allows premature termination to return a nearly optimal, feasible point

Barrier functions are implemented similar to penalties but must meet the following conditions:

1) Continuous
2) Non-negative
3) Approach infinity as x approaches boundary

Methods

- Inverse Barrier:

$$p_{\text{barrier}}(x) = -\sum_i \frac{1}{g_i(x)}$$

- Log Barrier

$$p_{\text{barrier}}(x) = -\sum_i \begin{cases} \log(-g_i(x)) & g_i(x) \leq -1 \\ 0 & \text{otherwise} \end{cases}$$